

6.3.2 Electrical characteristics

Table 8 Electrical Characteristics: Step down pre regulator

$V_{VS} = 6.0 \text{ V to } 40 \text{ V}$, $T_j = -40^\circ\text{C to } +150^\circ\text{C}$, all voltages with respect to ground, positive current flowing into pin (unless otherwise specified)

Parameter	Symbol	Values			Unit	Note / Test Condition	Number
		Min.	Typ.	Max.			
Step Down Pre Regulator							
Output voltage	$V_{\text{PREREG,BUCK}}$	5.65	5.8	5.95	V	PWM-mode; $V_{\text{VS}} \geq 6.5 \text{ V}$	P_6.3.2.1
Output voltage	$V_{\text{PREREG,BUCK}}$	5.6	5.8	6.00	V	PFM-mode; $V_{\text{VS}} \geq 6.5 \text{ V}$	P_6.3.2.2
Power stage high side switch on resistance	$R_{\text{ON, HS}}$	150	360	630	mΩ	$V_{\text{VS}} \geq 6 \text{ V}$	P_6.3.2.3
Power stage low side switch on resistance	$R_{\text{ON, LS}}$	50	200	300	mΩ	$V_{\text{VS}} \geq 6 \text{ V}$	P_6.3.2.4
Buck peak over current limit	$I_{\text{PREREG, max}}$	1.578	1.857	2.135	A	—	P_6.3.2.5
SW rise time	$t_{\text{R, Buck}}$	2	6	14	ns	¹⁾ $6.5\text{V} \leq V_{\text{VS}} \leq 18 \text{ V};$ $I_{\text{PREREG}} \geq 0.5 \text{ A}$	P_6.3.2.6
SW fall time	$t_{\text{F, Buck}}$	2	9	18	ns	¹⁾ $6.5\text{V} \leq V_{\text{VS}} \leq 18 \text{ V};$ $I_{\text{PREREG}} \geq 0.5 \text{ A}$	P_6.3.2.7
Maximum duty cycle	$D_{\text{BUCK, max}}$	-	-	100	%	—	P_6.3.2.8
Minimum switch on time	$t_{\text{ON, min}}$	20	50	80	ns	$I_{\text{PREREG}} \geq 0.5 \text{ A}$	P_6.3.2.9
Soft start ramp	$t_{\text{SS, BUCK}}$	70	190	380	μs	¹⁾ $V_{\text{PRE_REG,BUCK}}$ rising from 5% to 95% of $V_{\text{PREREG,nominal}}$; 2.2 MHz switching frequency, no load	P_6.3.2.10
Soft start ramp	$t_{\text{SS, BUCK}}$	0.7	2.0	3.5	ms	¹⁾ $V_{\text{PRE_REG,BUCK}}$ rising from 5% to 95% of $V_{\text{PRE_REG,nominal}}$; 400 kHz switching frequency, no load	P_6.3.2.11
Current threshold for transition from PWM to PFM	$I_{\text{PWM/PFM}}$	26	57	90	mA	—	P_6.3.2.12
Current threshold for transition from PFM to PWM	$I_{\text{PFM/PWM}}$	100	145	190	mA	—	P_6.3.2.13
Over temperature warning threshold	$T_{\text{J,OT, WRN}}$	130	145	160	°C	T_{J} increasing ¹⁾	P_6.3.2.14

Table 8 Electrical Characteristics: Step down pre regulator (cont'd)

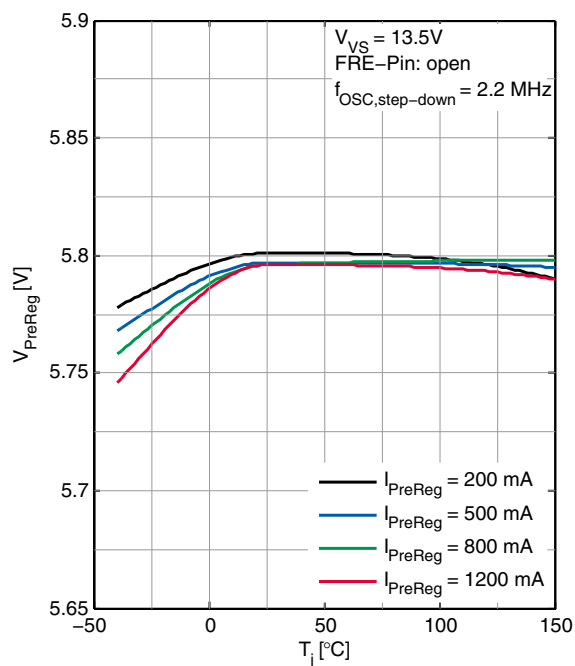
V_{VS} =6.0 V to 40 V, T_j = -40°C to +150°C, all voltages with respect to ground, positive current flowing into pin (unless otherwise specified)

Parameter	Symbol	Values			Unit	Note / Test Condition	Number
		Min.	Typ.	Max.			
Over temperature shutdown threshold	$T_{j,OT, shutdown}$	175	190	205	°C	T_j increasing 1)	P_6.3.2.15
Over temperature sensor hysteresis	$T_{j,OT, hyst}$	—	10	—	°C	1)	P_6.3.2.16

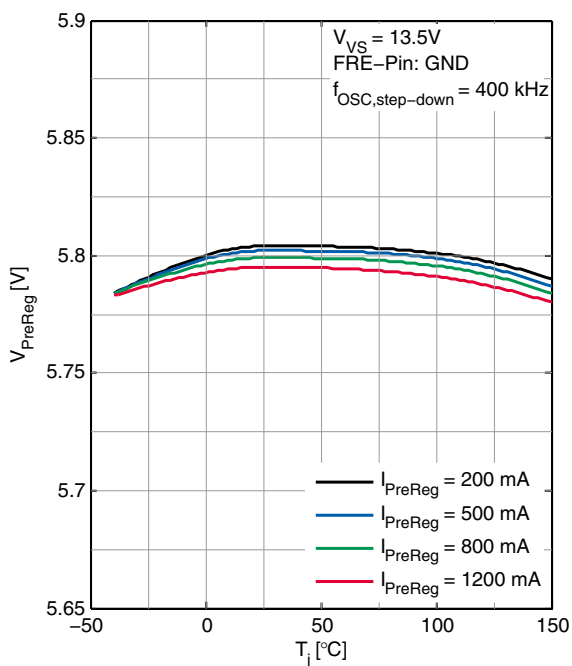
1) Specified by design, not subject to production test

6.3.3 Typical Performance Characteristics

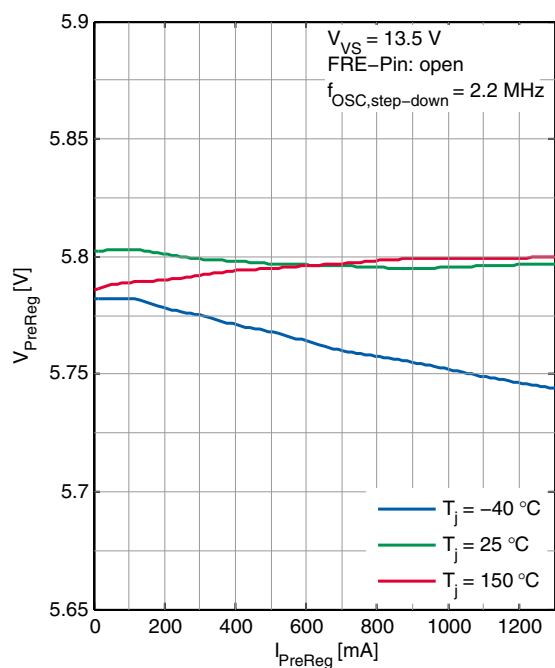
Pre Regulator Output Voltage V_{PreReg} versus Junction Temperature T_j (FRE-Pin: open)



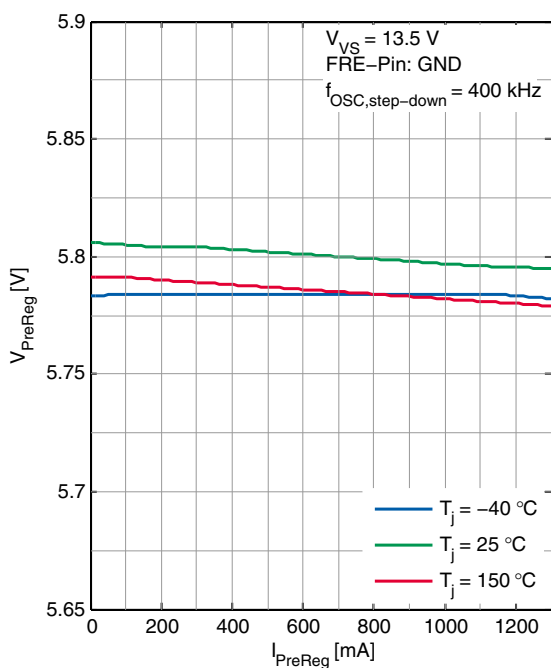
Pre Regulator Output Voltage V_{PreReg} versus Junction Temperature T_j (FRE-Pin: GND)



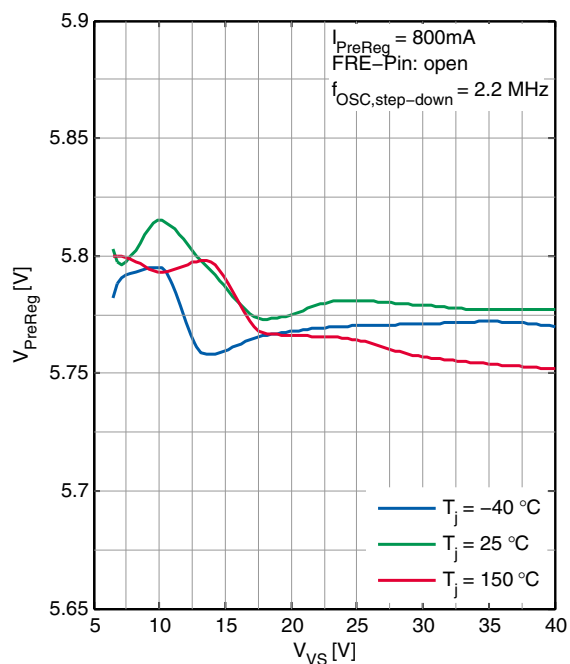
Pre Regulator Output Voltage V_{PreReg} versus Load Current I_{PreReg} (FRE-Pin: open)



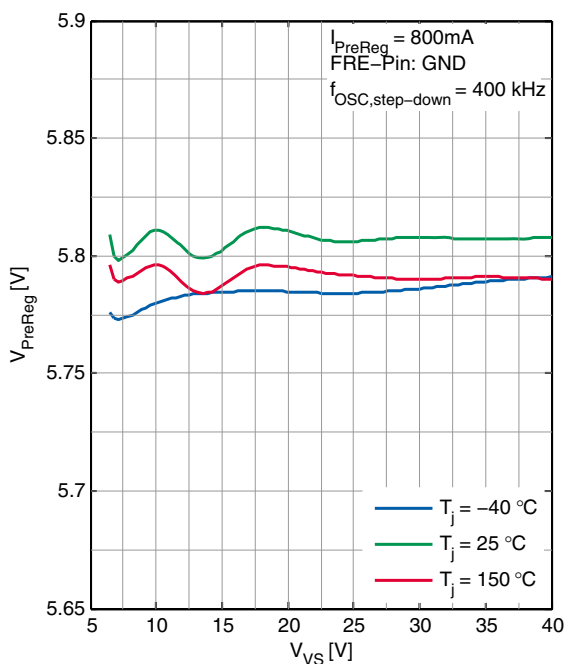
Pre Regulator Output Voltage V_{PreReg} versus Load Current I_{PreReg} (FRE-Pin: GND)



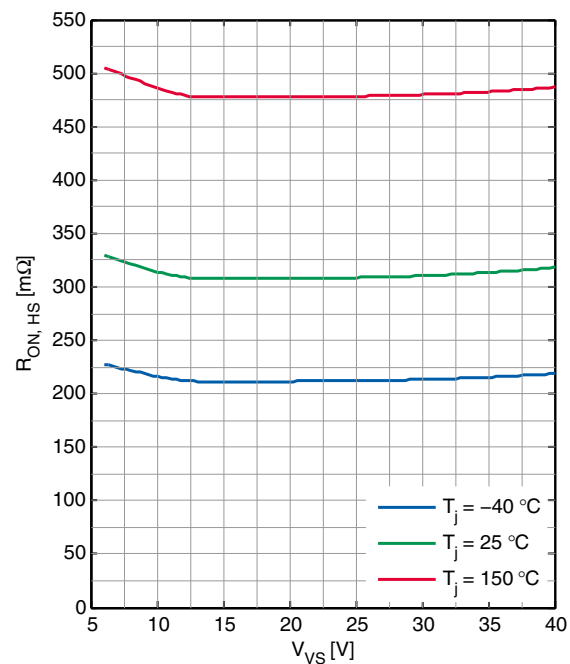
Pre Regulator Output Voltage V_{PreReg} versus Supply Voltage V_{VS} (FRE-Pin: open)



Pre Regulator Output Voltage V_{PreReg} versus Supply Voltage V_{VS} (FRE-Pin: GND)



High-Side Switch ON resistance $R_{\text{ON, HS}}$ versus Supply Voltage V_{VS}



Low-Side Switch ON resistance $R_{\text{ON, LS}}$ versus Supply Voltage V_{VS}

